

ECOM30001/90001 Basic Econometrics

Tutorial 9: Omitted Variable Bias & Instrumental Variables (IV)

Zheng Fan

Department of Economics, The University of Melbourne

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Tutorial 9 Overview

Today's Core Goals

- **Omitted Variable Bias (OVB):** Formula ($\hat{\beta}_y - \hat{\delta}_y = \hat{\gamma}_{k|y} \hat{\delta}_k$) and signing bias.
- **Endogeneity & IV:** Why education is endogenous and what makes an IV valid.
- **2SLS Estimation in R:** First-stage F -test for weak IVs and `ivreg()`.
- **Return to Education Puzzle:** Comparing $\hat{\beta}_{IV}$ vs. $\hat{\beta}_{OLS}$, plus Exam MCQ.

Question 1: Omitted Variable Bias (OVB)

The OVB Formula from Lectures

Suppose the short regression omits variable k . The bias on regressor y is:

$$\hat{\beta}_y - \hat{\delta}_y = \hat{\gamma}_{k|y} \cdot \hat{\delta}_k$$

where $\hat{\beta}_y$ is the short regression estimate, $\hat{\delta}_y, \hat{\delta}_k$ are long regression estimates, and $\hat{\gamma}_{k|y}$ is the slope of the auxiliary regression of k on y .

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Application: Production Function (Omitting $\log K$):

- **Largest parameter difference:** $\log(Y)$ has the largest change:

$$\hat{\beta}_y - \hat{\delta}_y = -0.238 - 0.569 = -\mathbf{0.807} \quad (\text{vs. } -0.676 \text{ for } \log(w/r))$$

- **Signing the correlation ($\hat{\gamma}_{k|y}$):**

$$-0.807 = \hat{\gamma}_{k|y} \times (-1.062) \implies \hat{\gamma}_{k|y} > 0$$

- **Economic Logic:** Producing more output (Y) requires more capital (K). Omitting $\log K$ forces its negative effect into $\log Y$, turning $\hat{\beta}_y$ negative in the short model.

Question 2: Empirical Wage Model

Econometric Specification (Dataset: wages9.csv, $N = 2,008$)

$$\begin{aligned}\ln(\text{wage}_i) = & \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{exper}_i^2 + \beta_4 \text{disadv}_i \\ & + \beta_5 \text{city}_i + \beta_6 \text{city10}_i + \sum_{j=2}^4 \beta_{7j} \text{region}_{ji} + \varepsilon_i\end{aligned}$$

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Key Parameter Interpretations:

- **Return to Education (β_1):** $100\beta_1\%$ is the percentage wage increase associated with an extra year of education, ceteris paribus.
- **Disadvantaged Area Penalty (β_4):** $100\beta_4\%$ is the approximate percentage wage differential for workers living in disadvantaged regions relative to non-disadvantaged regions, ceteris paribus.

Question 2(c): OLS Estimates & Historical Location Test

Regressor	Estimate	Robust SE	<i>t</i> value	Pr(> <i>t</i>)
(Intercept)	4.6730	0.0828	56.46	< 0.001
educ (β_1)	0.0751***	0.0042	17.82	< 0.001
exper	0.0820***	0.0079	10.42	< 0.001
exper ²	−0.0020***	0.0004	−5.25	< 0.001
disadv (β_4)	− 0.1183**	0.0361	−3.28	0.0011
city	0.1217***	0.0220	5.54	< 0.001
city10	0.0706**	0.0217	3.25	0.0012
region2	0.0521*	0.0222	2.34	0.0191
region3	0.0513	0.0393	1.31	0.1916
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Joint Hypothesis Test on Location 10 Years Ago

$H_0 : \beta_{\text{city10}} = \beta_{\text{region2}} = \beta_{\text{region3}} = \beta_{\text{region4}} = 0 \implies \mathbf{F(4, 1998)} = \mathbf{3.6616}$, $\mathbf{p = 0.0056}$.

► **Conclusion:** Reject H_0 . Past location has a persistent, statistically significant impact on current wages!

Question 2(d): The Endogeneity of Education

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The Economic Mechanism:

- **Omitted Ability in ε_i :** Natural talent, cognitive skill, grit, and motivation directly increase market wages.
- **Correlation with Schooling:** High-ability individuals find school easier and less costly, obtaining more education ($\text{Cov}(\text{educ}, \text{ability}) > 0$).
- **Consequence for OLS:** Standard OLS confuses the causal return to schooling with the return to innate ability $\implies$ **OLS is biased and inconsistent!**

Instrumental Variables: The Two Core Conditions

What makes a variable Z a valid instrument for endogenous X ?

To identify the causal effect of X on Y , an instrument Z must satisfy:

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Z must be correlated with endogenous education.

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Proposed Instruments: Mother's education (`meduc`) & Father's education (`feduc`).

Question 2(e): Step 1 — First-Stage Regression & Weak IV Test

First-Stage Model for Education

$$\text{educ}_i = \pi_0 + \pi_1 \text{meduc}_i + \pi_2 \text{feduc}_i + \mathbf{x}'_{1i} \gamma + v_i$$

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Instrument	Estimate	Robust SE	<i>t</i> value	Pr(> <i>t</i>)
meduc (π_1)	0.1575***	0.0210	7.49	< 0.001
feduc (π_2)	0.1901***	0.0189	10.05	< 0.001

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Testing Instrument Relevance (Rule of Thumb: $F > 10$)

Joint hypothesis test $H_0 : \pi_1 = 0, \pi_2 = 0$:

$$F(2, 1997) = 85.47 \gg 10 \quad (p < 0.001)$$

► **Conclusion:** Parental education variables are **strongly relevant** instruments!

Table 1: OLS vs. IV (2SLS) Wage Regressions

Regressor	OLS	IV (2SLS with <code>meduc</code> , <code>feduc</code>)
(Intercept)	4.6730*** (0.0828)	4.0162*** (0.2448)
educ	0.0751*** (0.0042)	0.1139*** (0.0142)
<code>exper</code>	0.0820*** (0.0079)	0.0989*** (0.0102)
<code>exper</code> ²	−0.0020*** (0.0004)	−0.0021*** (0.0004)
<code>disadv</code>	−0.1183** (0.0361)	−0.1203*** (0.0361)
<code>city</code>	0.1217*** (0.0220)	0.1088*** (0.0225)
<code>city</code> ₁₀	0.0706** (0.0217)	0.0661** (0.0220)
<code>region</code> ₂	0.0521* (0.0222)	0.0529* (0.0227)
<code>region</code> ₃	0.0513 (0.0393)	0.0663 ⁺ (0.0395)
<code>region</code> ₄	0.0338 (0.0285)	0.0206 (0.0294)
Observations	2,008	2,008
R^2	0.236	0.200

Question 2(f): The Return to Education Puzzle

Comparison of Return to Education

$$\hat{\beta}_{\text{OLS}} = 7.51\% \quad \text{vs.} \quad \hat{\beta}_{\text{IV}} = 11.39\%$$

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$$\hat{\beta}_{OLS} = 7.51\% \quad \text{vs.} \quad \hat{\beta}_{IV} = 11.39\%$$

Discussion: Is this compatible with Omitted Ability Bias?

NO! The standard omitted ability story predicts:

$$\text{Cov}(\text{educ}, \text{ability}) > 0 \implies \text{OLS is biased upwards} \implies \hat{\beta}_{OLS} > \beta_{\text{true}}$$

If IV purged ability bias, we would expect $\hat{\beta}_{IV} < \hat{\beta}_{OLS}$. But here $\hat{\beta}_{IV} > \hat{\beta}_{OLS}$!

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Why is $\hat{\beta}_{IV}$ higher than $\hat{\beta}_{OLS}$?

- ❶ **Invalid Instruments (Heritability):** Educated parents transmit innate ability to children $\implies \text{Cov}(\text{meduc}, \varepsilon) > 0$, adding upward bias to IV!
- ❷ **Measurement Error:** Reporting error in schooling causes attenuation bias in OLS.
- ❸ **LATE:** Returns are higher for individuals induced into college by parental support.

Exam Practice Question (MCQ)

Question: Identifying Valid Instruments

In the model $Y_i = \beta_0 + \beta_1 X_{1i} + \varepsilon_i$ with endogenous X_{1i} , candidate IVs are Z_{1i} and Z_{2i} :

$$E(X_{1i} \mid Z_{1i}, Z_{2i}) = \pi_0 + \pi_1 Z_{1i} + \pi_2 Z_{2i}$$

$$E(\varepsilon_i \mid Z_{1i}, Z_{2i}) = \gamma_0 + \gamma_1 Z_{1i} + \gamma_2 Z_{2i}$$

If $\pi_1 \neq 0$, $\pi_2 \neq 0$, $\gamma_1 \neq 0$, and $\gamma_2 = 0$, then:

- ① both Z_{1i} and Z_{2i} are valid IVs for X_{1i}
- ② only Z_{1i} is a valid IV for X_{1i}
- ③ only Z_{2i} is a valid IV for X_{1i}
- ④ none of the above

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Suggested Answer: Option 3 (only Z_{2i} is valid)

- **Relevance:** $\pi_1 \neq 0$ and $\pi_2 \neq 0 \implies$ both Z_1 and Z_2 are correlated with X_1 .
- **Exogeneity:** $\gamma_1 \neq 0 \implies Z_1$ is correlated with ε (**invalid/endogenous**).
 $\gamma_2 = 0 \implies Z_2$ is uncorrelated with ε (**exogenous/valid**).

Looking Ahead: Non-Teaching Week

Mid-Semester Break

- **Next Week: Non-Teaching Week** (No tutorials or lectures).
- **Classes Resume:** We will meet again in **two weeks** for Tutorial 10.

Enjoy the mid-semester break, and see you in two weeks!